

# Problem Set 5

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## 1 Questions and Responses

### Scraping

This data tracks grant awards from the Texas Space Commission. I thought it would be interesting to pair with federal data and estimate the effects of public R/D on a variety of outcomes similar to the Embrapa seminar last fall (Ackerman et al. 2026). I followed Grant McDermott’s SelectorGadget notes.

### API

I had to use the following packages: blackmarbler, geodata, sf, terra, ggplot2, tidyterra, lubridate, tidyverse, purrr. The table gives summary stats for Q1 in 2021. A potential useful piece of information is that average luminosity from February to March 2021 declined, and that the sample size declined between the two months.

Table 1: Monthly Summary of Nighttime Lights

| Month         | N    | Admin Units | Days | Mean | SD   | Min    | Max  |
|---------------|------|-------------|------|------|------|--------|------|
| January 2021  | 1612 | 52          | 31   | 1.86 | 4.44 | 0.0172 | 35.1 |
| February 2021 | 1456 | 52          | 28   | 1.84 | 4.41 | 0.0127 | 29.9 |
| March 2021    | 1612 | 52          | 31   | 1.71 | 3.98 | 0.0192 | 32.1 |